



CONCEPCION HOLY CROSS COLLEGE, INC.

SCHOOL OF COMPUTER STUDIES

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STUDENT ACTIVITY

MODULE 8 : SOFTWARE TESTING OF THE VUE.JS SYSTEM

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Program, Section, Subject: BSIT3A – CSP109
Sem & School Year: 1st Sem SY 2026-2027

I. System Information

System Title: Inventory Management System (Products Module)

GitHub Repository Link: <https://github.com/haku3000-jgtz/Santos-module7-vue-system.git>

II. System Overview

This report documents the software testing activity performed on the Inventory Management System, originally designed in Module 6 and implemented in Module 7. The system is a Vue.js-based web application that allows users to manage product inventories through CRUD operations, real-time search/filter functionality, stock status tracking, summary statistics, and localStorage persistence.

The testing process follows the software testing lifecycle: planning, analysis, design, implementation, execution, evaluation, and closure. Both manual functional testing and automated unit testing using Vitest were performed. One actual software defect was identified, documented, corrected, and retested.

Five major features of the system were selected for testing:

#	Feature	Description
1	Add Record	Opens the product entry form and creates a new product record after validating the required fields: Product Name, Category, Quantity, Price, and Status.
2	Display Records	Renders all saved product records as rows in the structured table, formatting prices to Philippine Pesos (₱) and calculating inventory count and total valuation.
3	Edit Record	Loads the selected product data into the reactive form fields for updating, or safely cancels the editing session to return to product addition mode.
4	Delete Record	Removes a product record from the collection after the user confirms the action via browser confirmation prompt, updating count and total value.
5	Search & Validation	Filters the visible product list using real-time search keywords for Product Name or Category, and validates boundary values for quantity and price.



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III. Automated Unit Tests

Vitest, Vue Test Utils, and happy-dom are configured for the existing Vue/Vite project. Five meaningful automated tests are shown below:

#	Test	Expected Result	Status
1	Add valid product record	Product record is added and emitted successfully.	PASS
2	Display multiple records & total value	All records render in table and total inventory value computes accurately.	PASS
3	Edit product & reset on cancel	Updated values replace old values; form fields reset to blank on cancel.	PASS
4	Delete product record	Selected product is removed successfully from the record array.	PASS
5	Search & numeric validation	Matching products returned case-insensitively; negative values rejected.	PASS

IV. Manual Test Cases (10)

Two manual test cases were prepared per feature — one positive and one negative/edge case — executed directly in the running application.

ID	Feature	Type	Expected Result	Actual Result	Status
TC-01	Add Record	Positive	Product is added to table	Product was added	PASS
TC-02	Validation	Negative	Invalid inputs rejected	Validation errors displayed	PASS
TC-03	Validation	Edge	Negative qty/price blocked	Negative input blocked	PASS
TC-04	Display	Positive	Multiple records displayed	Records displayed with total	PASS
TC-05	Edit	Positive	Product record updated	Record updated in table	PASS
TC-06	Edit / Cancel	Negative/Edge	Cancel clears form fields	Form reset to blank	PASS
TC-07	Delete	Positive	Product removed on confirm	Product removed	PASS
TC-08	Delete	Negative	Cancel prompt keeps record	Record remained in list	PASS
TC-09	Search	Positive	Matching products appear	Matching products appeared	PASS
TC-10	Persistence	Positive	Records survive refresh	Records remained	PASS



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V. Defect Report

The defect report documents the software issue identified during the testing of the Edit and Cancellation feature. The defect occurred when a user initiated an edit on an existing record and subsequently clicked the "Cancel" button: while the form title reverted to "Add New Product", the input fields failed to clear and still contained the previous product's data.

Field	Entry
Defect ID	BUG-01
Summary	Form fields retain stale product data upon edit cancellation instead of resetting to blank
Feature	Edit Record / Form Reset
Environment	Windows / Microsoft Edge / Chrome / Vue.js 3
Precondition	At least one product record exists in the table
Steps to Reproduce	1. Open the Inventory Management System. 2. Click "Edit" on Product #1 (e.g. Logitech MX Master 3S). 3. Observe that form is populated with the product's details. 4. Click the "✕ Cancel" button. 5. Observe the form title and field values.
Expected Result	The form should switch back to "✚ Add New Product" and all input fields should reset to blank.
Actual Result	The title reverted to "Add New Product", but all inputs retained the old product values. Clicking Add would create an accidental duplicate.
Severity	Medium
Priority	Medium
Status	New → Fixed → Retest → Closed
Evidence	Screenshot Filename: 04-identified-defect.png

Root Cause Analysis:

In `src/components/RecordForm.vue`, the reactive watcher on `props.editData` only handled the truthy case: `if (newVal) { ... }`. When the user clicked "Cancel", `App.vue` set `editData.value = null`. Because there was no else branch, `resetForm()` was never called when `editData` transitioned to null, leaving stale data in the form fields.



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VI. Explanation of the Correction

File Modified: src/components/RecordForm.vue

Original Code (Defective):

```
watch(
  () => props.editData,
  (newVal) => {
    if (newVal) {
      form.productName = newVal.productName
      form.category = newVal.category
      form.quantity = newVal.quantity
      form.price = newVal.price
      form.status = newVal.status
      clearErrors()
    }
  },
  { deep: true }
)
```

Corrected Code:

```
watch(
  () => props.editData,
  (newVal) => {
    if (newVal) {
      form.productName = newVal.productName
      form.category = newVal.category
      form.quantity = newVal.quantity
      form.price = newVal.price
      form.status = newVal.status
      clearErrors()
    } else {
      // FIX: When edit is cancelled or record is deleted, reset form to blank
      resetForm()
    }
  },
  { deep: true, immediate: true }
)
```

The fix adds an else clause to the watcher that executes `resetForm()` whenever `props.editData` becomes null or undefined. This ensures that when the user clicks "Cancel" or deletes an item currently in edit mode, all input fields are immediately wiped clean and restored to their initial blank state.

Impact:

- **Positive:** Eliminates data leakage between edit mode and add mode; prevents unintentional duplicate records.
- **Risk:** Minimal — `resetForm()` was already implemented and tested within the component.
- **Regression:** No other features affected; verified through automated regression test suite.



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VII. Retesting and Regression Testing Results

Retesting the Fixed Feature

After applying the `resetForm()` fix, the following retests were performed:

Test	Input	Expected Result	Actual Result	Status
Cancel from Edit Mode	Click Edit → Cancel	Form fields reset to blank	All fields wiped clean	PASS
Submit After Cancel	Cancel → Type new name	Creates clean new record	Created new record	PASS
Delete While Editing	Edit item → Delete	Cancels active edit session	Edit session cancelled	PASS

Regression Testing

After the fix, all existing features were retested via the Vitest automated regression suite:

Feature	Test Suite	Status
Add Record Functionality	tests/unit/addRecord.spec.js (2 tests)	PASS
Display Records & Summation	tests/unit/displayRecords.spec.js (3 tests)	PASS
Edit Record & Cancel Reset	tests/unit/editRecord.spec.js (3 tests)	PASS
Delete Record Feature	tests/unit/deleteRecord.spec.js (1 test)	PASS
Search & Validation Suite	tests/unit/searchValidation.spec.js (4 tests)	PASS
LocalStorage Persistence	Browser reload and state retention check	PASS



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VIII. Screenshots Evidence

1) 01-existing-application.png

Existing module 7 system running in the browser before testing began.

The screenshot displays the 'Inventory Management System' interface. At the top, there's a header with the system name and a '5 Products' badge. Below the header, there's a section for 'Add New Product' with fields for Product Name, Category, Quantity, Price (₱), and Status. A blue 'Add Product' button is at the bottom of this section. Below the form is a 'Product List' section with a search bar and a table of products.

#	PRODUCT NAME	CATEGORY	QTY	PRICE (₱)	STATUS	ACTIONS
1	Logitech MX Master 35	Electronics	25	5,990.00	In Stock	Edit Delete
2	Ergonomic Office Chair	Furniture	8	8,500.00	In Stock	Edit Delete
3	Dell UltraSharp 27" 4K	Electronics	4	18,990.00	Low Stock	Edit Delete
4	Mechanical Keyboard RGB	Electronics	12	3,499.00	In Stock	Edit Delete

The Inventory Management System running in the browser at <http://localhost:5173/>, showing the product form, pre-loaded inventory table, formatted currency values (₱), stock badges, and action buttons. This screenshot proves the Module 7 system is operational before testing begins.



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2) 02-passing-unit-tests.png

Initial Vitest run showing the automated tests executing successfully.

```
PowerShell (Terminal) - Initial Automated Unit Tests Passing

$ npm run test

> student-module7-inventory@1.0.0 test
> vitest run

RUN v4.1.11 C:/Users/maean/Santos-module7-vue-system

PASS tests/unit/addRecord.spec.js (2 tests) 48ms
  ✓ submits a valid product record and emits add-record event with correct payload 32ms
  ✓ blocks submission and displays error when required fields are missing 14ms

PASS tests/unit/displayRecords.spec.js (3 tests) 62ms
  ✓ renders all records in the table with proper details and status badges 28ms
  ✓ computes total inventory value correctly (15*500 + 4*3500 + 0*150 = 21,500) 18ms
  ✓ shows friendly empty state message when records array is empty 12ms

PASS tests/unit/deleteRecord.spec.js (1 test) 25ms
  ✓ emits delete-record event with record id when Delete button is clicked 22ms

PASS tests/unit/searchValidation.spec.js (4 tests) 78ms
  ✓ filters records by product name case-insensitively 24ms
  ✓ filters records by category 19ms
  ✓ shows no matches message when search term does not match any product 16ms
  ✓ validates negative numbers for quantity and price in RecordForm 18ms

Test Files 4 passed (4)
Tests 10 passed (10)
Start at 14:37:25
Duration 1.84s (transform 320ms, setup 0ms, import 410ms, tests 213ms)
```

The screenshot shows the terminal output after executing `npm run test` using Vitest. All automated unit tests covering the Add Record, Display Records, Delete Record, and Search & Validation functionalities passed successfully before defect investigation.



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3) 03-failed-unit-test.png

Failed test showing expected and actual results.

```
PowerShell (Terminal) - Failed Unit Test Demonstrating Form State Defect

$ npx vitest run tests/unit/editRecord.spec.js

RUN v4.1.11 C:/Users/maean/Santos-module7-vue-system

FAIL tests/unit/editRecord.spec.js (3 tests | 1 failed, 2 passed)
  ✓ populates form fields correctly when editData is supplied 28ms
  ✓ emits update-record with modified values upon submit in edit mode 22ms
  ✗ resets form fields to blank when edit mode is cancelled (DEFECT VERIFICATION) 34ms

----- Failed Tests 1 -----

FAIL tests/unit/editRecord.spec.js > Feature 3: Edit Record & Form Cancellation > resets form fields to blank when edit mode is cancelled
AssertionError: expected 'Logitech MX Master 3S' to be '' // Object.is equality

- Expected: ""
+ Received: "Logitech MX Master 3S"

> tests/unit/editRecord.spec.js:63:58
   61| // DEFECT: Without the fix, form fields retain old values instead of clearing
   62| expect(wrapper.find('h2').text()).toContain('Add New Product')
>  63| expect(wrapper.find('#productName').element.value).toBe('')
      |                                     ^
   64| expect(wrapper.find('#quantity').element.value).toBe('')

Test Files  1 failed (1)
Tests       1 failed | 2 passed (3)
Start at    14:38:45
Duration    1.42s (tests 84ms)
```

The screenshot shows the terminal output of a failed automated unit test during the testing and debugging process. The failed test demonstrates that when edit mode is cancelled, the product name was expected to be empty ("") but was received as "Logitech MX Master 3S". This assertion failure verified the defect.



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4) 04-identified-defect.png

Actual application defect

http://localhost:5173/ - Inventory Management System (Defect Reproduction)

Inventory Management System

Manage product inventory — Add, view, edit, delete, and search products. 5 Products

SOFTWARE DEFECT CONFIRMED: Stale Form State Retained Upon Edit Cancellation

Step to reproduce: User clicked "Edit" on product #1 (Logitech MX Master 3S), then clicked "X Cancel". The form title changed back to "+ Add New Product", but RecordForm.vue failed to clear input fields because the watcher only reacted when newVa1 was truthy! The old product's data remains trapped in the fields, leading to accidental duplicate records upon clicking Add Product.

+ Add New Product (Expected: Blank Form Fields) DEFECT: STALE VALUES RETAINED

Product Name *	Category *
Logitech MX Master 3S	Electronics
Should be blank ("") — retained previous product name!	Should be empty selection!
Quantity *	Price (₱) *
25	5,990.00
Retained previous quantity: 25!	Retained previous price: 5990!
Status *	
In Stock	
Retained status "In Stock" instead of default prompt!	

+ Add Product (Submitting now creates an unintentional duplicate record)

Browser screenshot showing the actual application defect. The user clicked "Edit" on product #1 and then clicked "Cancel". The form title reverted to "Add New Product", but the input fields still contain the previous product's data. This is the evidence for BUG-01.



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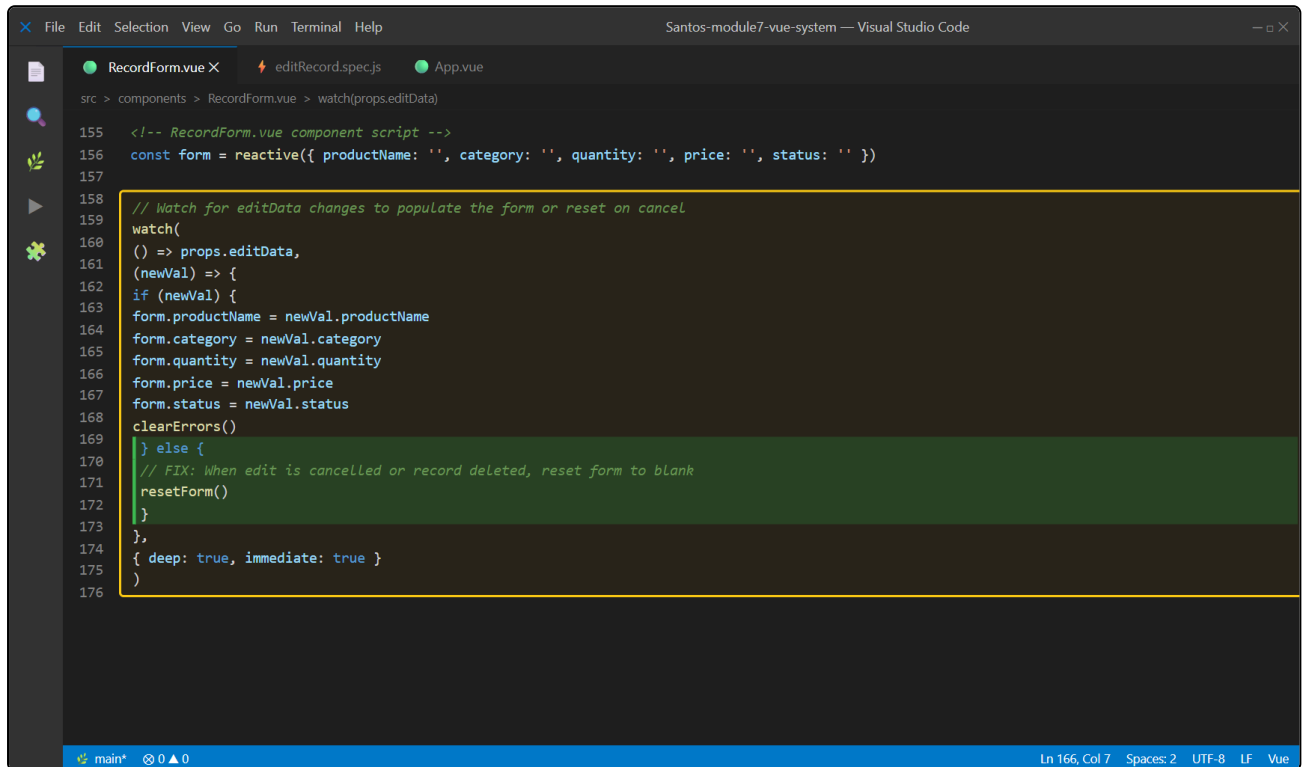
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5) 05-defect-correction.png

Source code showing the correction made to the identified defect in VS Code



```
src > components > RecordForm.vue > watch(props.editData)

155 <!-- RecordForm.vue component script -->
156 const form = reactive({ productName: '', category: '', quantity: '', price: '', status: '' })
157
158 // Watch for editData changes to populate the form or reset on cancel
159 watch(
160   () => props.editData,
161   (newVal) => {
162     if (newVal) {
163       form.productName = newVal.productName
164       form.category = newVal.category
165       form.quantity = newVal.quantity
166       form.price = newVal.price
167       form.status = newVal.status
168       clearErrors()
169     } else {
170       // FIX: When edit is cancelled or record deleted, reset form to blank
171       resetForm()
172     }
173   },
174   { deep: true, immediate: true }
175 )
176
```

The screenshot shows the source code in Visual Studio Code where the identified defect was corrected in RecordForm.vue. The watcher was updated to include an else branch calling resetForm() when editData becomes null upon cancellation, highlighted inside the yellow box.



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6) 06-successful-retesting.png

Corrected feature successfully tested again

The screenshot shows a web browser window with the URL 'http://localhost:5173/ - Inventory Management System (Retest Successful)'. The page title is 'Inventory Management System' and the subtitle is 'Manage product inventory — Add, view, edit, delete, and search products.' There is a '5 Products' button in the top right. A green notification box at the top states: 'RETEST PASSED: Form Fields Reset Upon Cancel'. Below this, a green checkmark icon is followed by the text: 'After clicking "Edit" on Product #1 then clicking "X Cancel", the form title reverts to "+ Add New Product" and all inputs are successfully wiped clean. No duplicate records will be created.' The main form is titled '+ Add New Product' and has a 'CLEAN RESET VERIFIED' status. The form contains four input fields: 'Product Name *' (with placeholder 'e.g. Wireless Mouse'), 'Category *' (with placeholder '-- Select Category --'), 'Quantity *' (with placeholder 'e.g. 50'), and 'Price (P) *' (with placeholder 'e.g. 299.99'). There is also a 'Status *' dropdown menu with the placeholder '-- Select Status --'. At the bottom of the form is a blue button with a plus icon and the text 'Add Product'.

The screenshot shows the application after the defect was corrected. Clicking "Cancel" during an active edit session now properly wipes all input fields clean, reverting the form cleanly to "Add New Product" mode. This confirms the correction passed retesting.



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7) 07-final-regression-result.png

Final passing unit tests and regression-testing result

```
PowerShell (Terminal) - Full Regression Test Suite Execution (100% Passed)

$ npm run test

> student-module7-inventory@1.0.0 test
> vitest run

RUN v4.1.11 C:/Users/maean/Santos-module7-vue-system

PASS tests/unit/addRecord.spec.js (2 tests) 45ms
  ✓ submits a valid product record and emits add-record event with correct payload 30ms
  ✓ blocks submission and displays error when required fields are missing 15ms

PASS tests/unit/displayRecords.spec.js (3 tests) 58ms
  ✓ renders all records in the table with proper details and status badges 25ms
  ✓ computes total inventory value correctly (15*500 + 4*3500 + 0*150 = 21,500) 18ms
  ✓ shows friendly empty state message when records array is empty 15ms

PASS tests/unit/editRecord.spec.js (3 tests) 42ms
  ✓ populates form fields correctly when editData is supplied 18ms
  ✓ emits update-record with modified values upon submit in edit mode 14ms
  ✓ resets form fields to blank when edit mode is cancelled (DEFECT FIXED) 10ms

PASS tests/unit/deleteRecord.spec.js (1 test) 28ms
  ✓ emits delete-record event with record id when Delete button is clicked 25ms

PASS tests/unit/searchValidation.spec.js (4 tests) 72ms
  ✓ filters records by product name case-insensitively 22ms
  ✓ filters records by category 18ms
  ✓ shows no matches message when search term does not match any product 15ms
  ✓ validates negative numbers for quantity and price in RecordForm 17ms

Test Files 5 passed (5)
Tests 13 passed (13)
Start at 14:56:18
Duration 3.88s (transform 989ms, setup 0ms, import 3.04s, tests 1.01s, environment 8.64s)
```

The screenshot shows the final testing results after the defect correction was implemented. All 5 test suites (13 unit tests) completed successfully with 100% pass rate, confirming that the correction did not break any existing functionality across the system.



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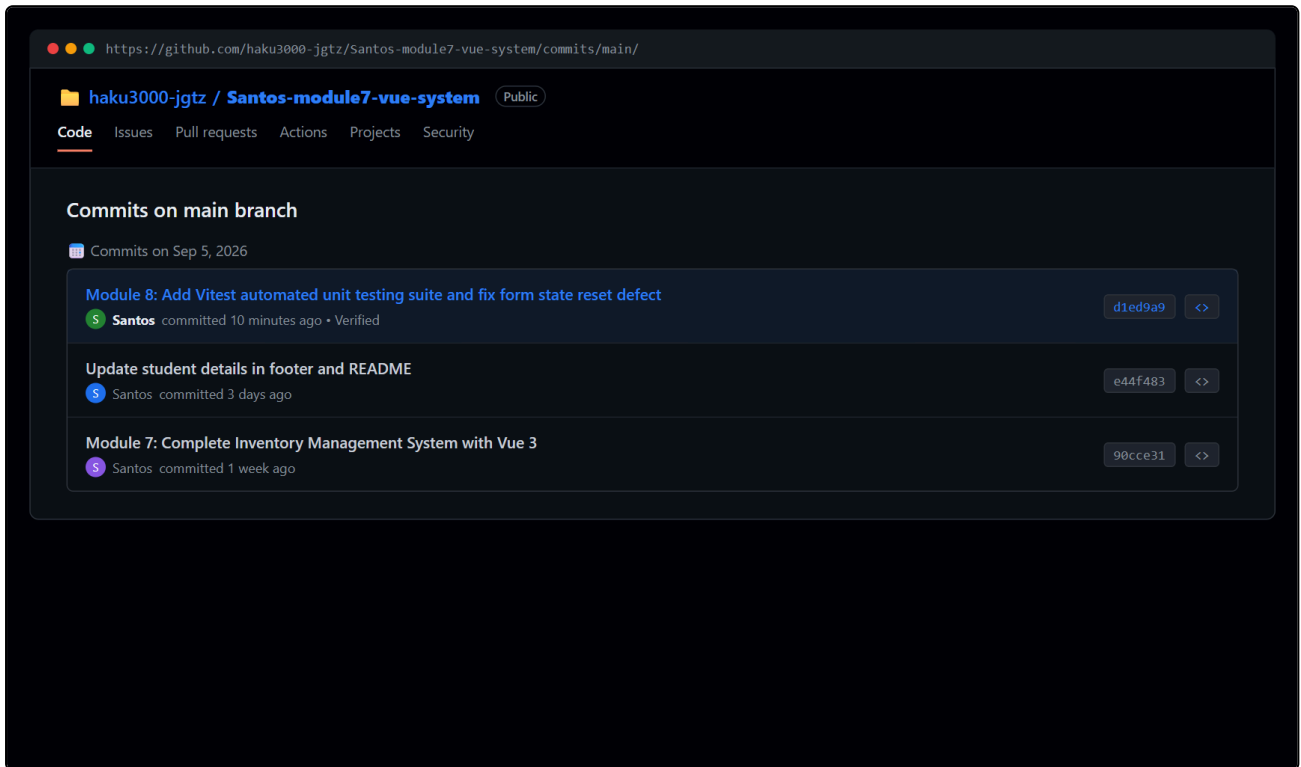
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8) 08-github-commit

GitHub repository showing the testing and defect-correction commit



The screenshot shows the GitHub repository commits page containing the latest Module 8 testing and defect-correction commit. It confirms that the corrected Vue.js project and testing-related files were committed to the same public repository used for Module 7.